1. H2 settled: genus-2 IC on glued h = 1 instances against negation rho

Run `run-20261019T1730Z` (decision run, seed tag 1), `run-20261019T1700Z` (run 1, failed control, retained). Protocol: `PROTOCOL.md` plus Amendment 1. Harness: `examples/h2_glue_cold.rs` (fork of 2170′s `n2_routes.rs`, blob `df0f097b…`). Host: shared 4-core cloud VM, no PMU; wall times are descriptive only.

1.1. Answer (measured)

With every cost charged cold and all 320 answers verified, glued genus-2 IC at $h = 1$ is **not cheaper than negation rho in any of the 12 price cells at any native price**. The ratio IC/rho ranges 1.00 to 2.20 (figure 1). By the frozen rule, 6 cells are *rho-wins*, 6 are *unresolved* (only because at $\nu = 0$, free native work, the ratio is 0.996 to 1.08 with an interval containing 1) and 0 are *IC-wins*. The frozen rule's branch is therefore "scope the 22M + I addition", not "close H2". The scoping (scoping section) shows that this addition would not change the verdict outside $\nu = 0$.

1.2. Design (frozen before runs)

16 glued instances with exact $h = 1$ (8 at $\log_2 N \approx 36$, 8 at 40), 20 targets each = 320 targets, all arms on every target. Arms: J (2170 `arm_glue`, double-large-prime IC, 41M + I per Jacobian addition on its sextic model), R (negation rho, default DP), RM (equal-memory rho: DP density chosen by a fixed memory rule), plus controls Rp (independent-seed A/A), Rr (identical-seed replay), Pl (plain rho, positive control). J is charged construction, both transfers, the sqrt-table build (p native ops), one Legendre test per degree-2 divisor priced at I, and all failed work. Prices: $\sigma = S/M \in \{0, 0.5, 1\}$, $\iota = I/M \in \{1, 10, 100, 1000\}$; native ops at $\nu \in \{0, 0.1, 1\}$ GAE each. Ratios are sums over matched targets; intervals are two-level (instance, target) bootstrap, 4000 resamples.

1.3. Controls (run 2)

Exact replay (Rr = R)	identical counters on all 320 targets
Replay pass	192 re-run rows identical in steps, M, S, I, native, words, Legendre
Independent A/A Rp/R	1.0225 to 1.0228 over 36 combinations; all 36 intervals contain 1
Positive control Pl/R	1.394 to 1.410; all intervals above 1
Rho calibration	steps / (0.886 $\sqrt{\ell}$) = 1.06 (36 bit), 1.02 (40 bit)
Verification	0 unverified rows of 4,032 (incl. replays)
Wall A/A	Rp/R seconds = 1.083: wall not qualified, none claimed

Run 1 (160 targets) passed all of these except my own point tolerance on the independent A/A (0.938, interval contains 1; tolerance $|r - 1| < 0.03$ was tighter than the sampling error of 160 rho walks). Under the frozen rule that run decides nothing; Amendment 1 replaced the tolerance by interval containment and fixed a fresh-seed replicate before it ran. Run 1′s cell ratios agree with run 2 within 0.06.

1.4. Result: cost ratio J / comparator

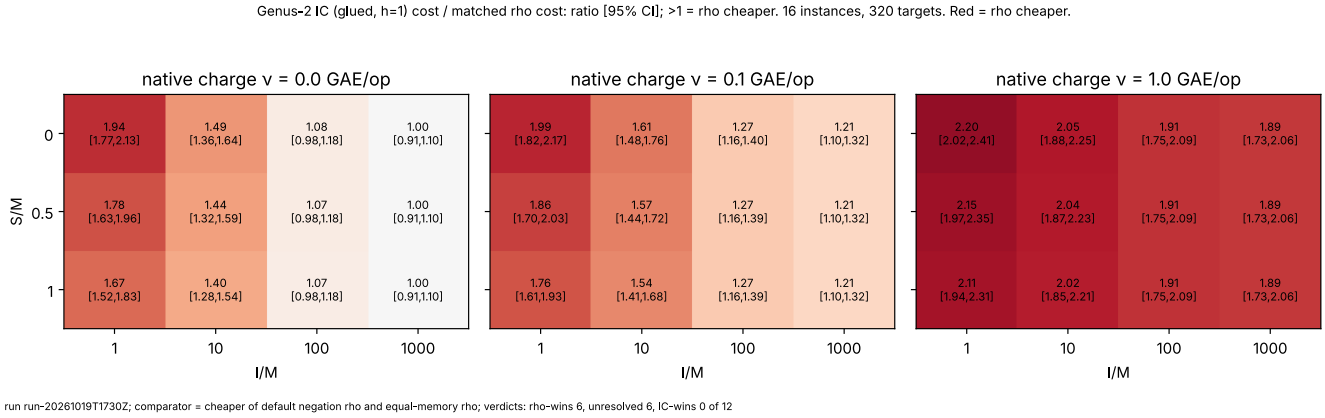


Figure 1: J / cheaper-of(R, RM) in each of 12 price cells, three native prices; ratio [95% interval]. Source: `results/run-20261019T1730Z/analysis.json`.

The backend's measured cost per Jacobian addition is $34M + 7S + 1I$ (= 41 M + I at $S = M$), against $9M + 4S + I$ for an $E(F_{p^2})$ addition (= 12 M + I at $S = 0.75M$), identical on all 16 instances. J takes 0.44 to 0.45 $\sqrt{\ell}$ steps (rho 0.91 to 0.94 $\sqrt{\ell}$): the count advantage is real (about 2.0×) but each J step is dearer: all charged work per step is 38.6 M + 8.3 S + 1.04 I + 7.4 native ops, against 9.2 M + 4.1 S + 1.02 I + 2.0 native for rho. Distance to the matched rho at $\nu = 0.1$: 1.20 (at $\iota = 1000$) to 1.99 (at $\iota = 1$, $\sigma = 0$).

1.5. Time × memory

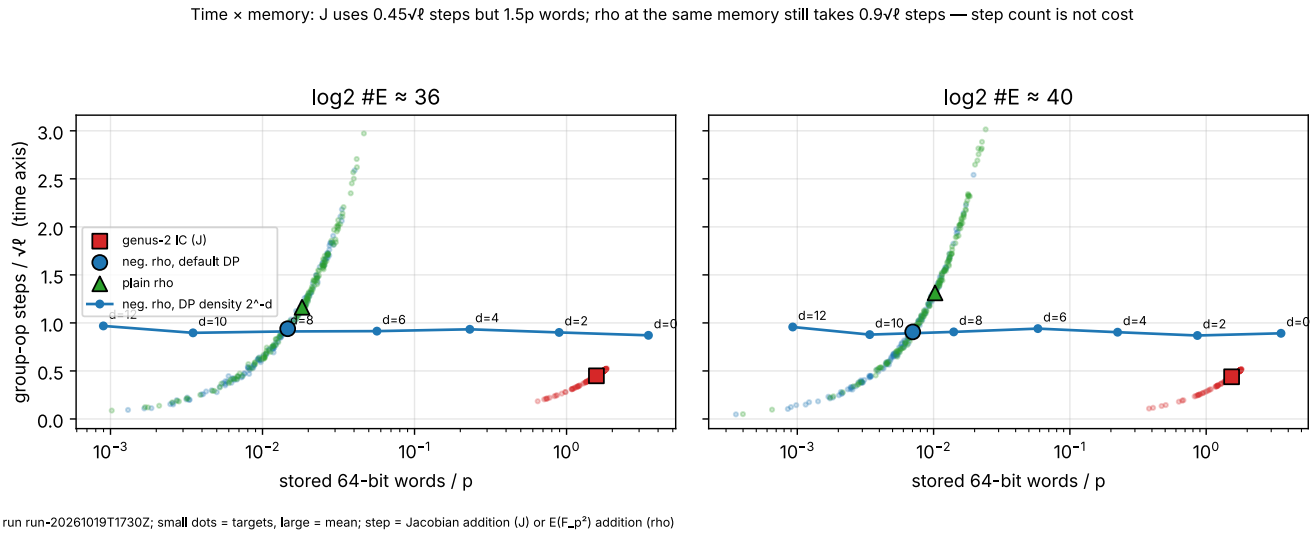


Figure 2: Steps/$\sqrt{\ell}$ against stored 64-bit words/p, per target (dots) and mean (markers). J stores 1.5 p words; negation rho at DP density 2^{-d}, from 0.001 p to 3.5 p words, stays at 0.88 to 1.0 $\sqrt{\ell}$ steps.

Memory does not buy rho speed beyond removing DP overhead, so rho at J's memory (RM, $d^* = 2$ on all 16 instances, 0.89 p words) is about as cheap as default rho and J's Θ(p) memory is a cost with no counterpart: J holds about 100 to 200× more words than default rho for a 2× step advantage that its per-step cost erases. A memory-capped J degenerates into distinguished-point rho (2170 H4); not rerun.

1.6. Scoping the ≤ 22M + I addition (frozen rule's branch)

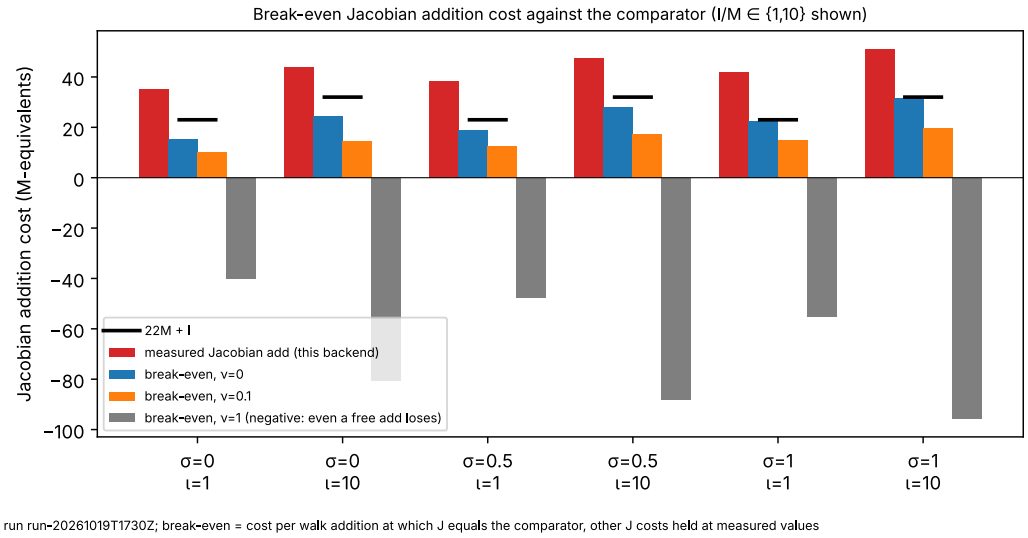


Figure 3: Per-addition cost at which J would equal the comparator (other J costs measured). Bars at $\nu = 1$ are negative.

- At $\nu = 0$ break-even is about 22 M + I (15 to 22 M-equivalents at $\iota = 1$, i.e. 2170's figure), so 2170's number is the free-native break-even.
- At $\nu = 0.1$ break-even is 10 to 65 M-equivalents, i.e. below 22 M + I in every cell (e.g. 14.6 at $\iota = 10, \sigma = 0$ against 32). Even a hypothetical 22 M + I addition leaves J at 1.20 to 1.51 × the comparator.
- At $\nu = 1$ break-even is negative in every cell: J's native work (7.4 ops/step) alone exceeds rho's total cost; no addition formula can win.
- Hypothetical column (22 M + I, not built): rho-wins 3, unresolved 9, IC-wins 0.
- The Legendre pricing does not carry the result: pricing it at 0 or 30 M gives the same tally (6 / 0 / 6), although at $\nu = 0.1$ the ratio dips to 0.81 under the Legendre = 0 variant in at least one cell; IC-wins stays 0 of 12.

Not built. The frozen rule said "scope and build". The data above show that the build target cannot flip any cell outside $\nu = 0$, so building it would change no verdict; this is a judgement the owner can override. The literature bound for real-model genus-2 addition (Erickson–Jacobson–Stein 2011, ePrint 2011/105) was not retrieved here, so no claimed operation count for it is made.

1.7. Verdict and limits

- H2 as stated (2170: IC beats negation rho on h = 1 glued instances) is **not supported**: 0 of 12 cells favour IC; 6 are rho-wins, 6 are unresolved (ties at $\nu = 0$ only). The rule's literal branch is "scope", done; the data say the next lever is native work and memory, not the addition formula.
- Next improvement to try: cut J's native work and $\Theta(p)$ memory (7.4 native ops and 1.6 p words per step), which dominates at $\nu \geq 0.1$; only then does the addition cost matter.
- Scope: 36 and 40 bit group sizes, one construction (2170′s gluing), cost-model counts (no hardware speed). 2229′s ecbench genus adapter accepts only the Scholten family and a 70M + 3S + I backend, so its adapters were not run; this study re-implements 2229′s protocol rules in a standalone driver. No ecbench session, leaderboard row or retired-instruction metric exists (no PMU).
- No improvement over the IC baseline is claimed (null result); leaderboard unchanged.